

The Generative Content of a Conserved Emptiness

Kinematic Voids as Superselection Generators:
the Salem Slot and the Five Objects Its Charge Produces

A Companion to *Lehmer's Box*, *The Emission-Gap Theorem*,
and *The Occupant of the Salem Slot*

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Abstract

A forbidden region can be empty in two ways. A *dynamic* void is a barrier the system could cross but pays not to; such voids leak, and the leak is the interesting object. A *kinematic* void has zero weight in the algebra the operations act on, by a conserved charge; nothing approaches it, and the enforcing law never fires. We argue a kinematic void is not inert: *kinematic emptiness* $\equiv$ *a conserved charge* $\equiv$ *a superselection rule*, and a conserved charge produces structure. On a minimal worked instance — the Salem slot, whose charge confines every root's argument to $\frac{\pi}{2}\mathbb{Z} \cong \mathbb{Z}/4\mathbb{Z}$ — we exhibit **five canonical objects, all verified** in exact arithmetic: (I) a $\mathbb{Z}/4\mathbb{Z}$ *grading* of the algebra, the operators acting as add ($\otimes$), double (squaring), union ($\oplus$); (II) a *content polynomial* $x^4 - 1 = \Phi_1\Phi_2\Phi_4$ governing all on-circle structure; (III) the *gap*, the cost floor $M \in \{1\} \cup [\varphi, \infty)$; (IV) a graded *normal form*, cyclotomic times grow; (V) a *conserved current*, the charge as a dynamical invariant. We then state the load-bearing consequence directly: the floor φ is *derived*, not *posited*. The framework fixes only the decision arity — growth decisions are ternary, hence degree two — and φ falls out the far end as the minimum realizable growth. The chain $\text{degree } 2 \Rightarrow \mathbb{Z}/4\mathbb{Z} \Rightarrow \varphi$ is three forced minima producing three forced outputs; the golden ratio is a consequence of the decision structure, not a constant chosen for being golden. We close on scope: the equivalence is domain-universal — Pauli exclusion, charge superselection, and type preservation are the same skeleton — while this instance generates the five objects *within* its own number theory. Same skeleton, different content; no output is exported across domains.

1 The substrate: a conserved emptiness is a generator

Two regions can both be empty for opposite reasons, and the distinction is the whole point.

Definition 1.1 (Dynamic vs. kinematic emptiness). A region of state space is *dynamically empty* for a process if the states exist in the operative algebra but are reached only at a cost the process avoids; such regions are approached, tunnelled into, or weakly populated (an energy barrier; a “forbidden” spectral line). A region is *kinematically empty* if its states carry zero amplitude in the algebra the operations act on, enforced by a quantity those operations preserve; nothing approaches it, and the enforcing law never has to act.

The thesis is the chain

kinematic emptiness $\equiv$ a conserved charge $\equiv$ a superselection rule
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together with the claim that the rightmost object is *productive*. A superselection charge is never merely a prohibition; it is a structure-building datum. In the abstract it furnishes: a *grading* of the algebra by charge value; a distinguished *content polynomial* whose roots are the allowed boundary states; a *gap* (the forbidden values become a quantitative bound); a *normal form* (objects factor along the grading); and a *conserved current* (the charge is constant under any admissible flow). This paper takes a minimal kinematic void — one where every step is decidable in exact arithmetic — and exhibits all five.

The void is the *Salem slot* of the companion papers: the regime “on the unit circle yet expanding” that the ternary growth decision $\text{spec}(\text{ad}_R) = \{-\sqrt{5}, 0, +\sqrt{5}\}$ has no channel for. The companion note *The Occupant of the Salem Slot* showed that nothing inhabits it, that a would-be occupant is expelled by the trace map $\beta \mapsto \beta + \beta^{-1}$, and that entry is blocked by a conserved charge: the confinement of every root’s argument to $\frac{\pi}{2}\mathbb{Z}$. That charge is the generator; the five objects below are its output.

One word, one meaning. We use “generation” in a single sense: the production of the five canonical objects of §§2–6, each a theorem with a verification attached. The closing claim that refusing to be filled is the act of generation refers to exactly these five — the grading, the content polynomial, the gap, the normal form, the conserved current — and to nothing wider.

2 Object I: the $\mathbb{Z}/4\mathbb{Z}$ grading

Definition 2.1 (The angle charge). For $P \in \mathbb{Z}[x]$ define the *angle support* as the multiset of arguments of its roots, taken in $\mathbb{R}/2\pi\mathbb{Z}$. Say P carries charge in $\mathbb{Z}/4\mathbb{Z}$ if every argument lies in $\frac{\pi}{2}\mathbb{Z} = \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$; the charge of a root at argument $k\frac{\pi}{2}$ is $k \bmod 4$. The emission algebra $\mathcal{S}$ is generated by the seed catalog under $\oplus$ (direct sum), $\otimes$ (tensor), and $(\cdot)^2$ (squaring).

Theorem 2.2 (The grading). *Every object of $\mathcal{S}$ carries charge in $\mathbb{Z}/4\mathbb{Z}$, and the operators act on the charge by*

$$\otimes : \text{add} \quad (\arg \lambda\mu = \arg \lambda + \arg \mu), \quad (\cdot)^2 : \text{double} \quad (\arg \lambda^2 = 2 \arg \lambda), \quad \oplus : \text{union},$$

all internal to $\mathbb{Z}/4\mathbb{Z}$ since it is closed under $+$, $\times 2$, and union. The full group is realized: the real seeds carry $\{0, 2\}$, and the quartic generator $K = x^4 + 5x^2 - 5$ carries $\{0, 1, 2, 3\}$ (its imaginary pair $\pm i\beta$ sits at $\pm \frac{\pi}{2}$). Hence $\mathcal{S}$ is a $\mathbb{Z}/4\mathbb{Z}$ -graded structure. [COMPUTED] (verified on the catalog and on $\otimes, (\cdot)^2$ closures; charge multisets $\varphi:\{0, 2\}$, $\varphi\otimes\varphi:\{0, 0, 2, 2\}$, $\varphi^2:\{0, 0\}$, $K:\{0, 1, 2, 3\}$)

This is the generator itself, exhibited as the first object: an order-four conserved charge, with the three operators realizing the group law (addition, doubling, union) on $\mathbb{Z}/4\mathbb{Z}$. The kinematic content of the slot is now explicit — a Salem number has an on-circle conjugate at an *irrational* multiple of π , charge $\notin \mathbb{Z}/4\mathbb{Z}$, so the Salem regime is the off-charge sector, and no finite word in the operators leaves $\mathbb{Z}/4\mathbb{Z}$. The remaining four objects are read off this grading.

3 Object II: the content polynomial $x^4 - 1$

Proposition 3.1 (The content polynomial). *All on-circle roots of objects in $\mathcal{S}$ are fourth roots of unity — roots of*

$$x^4 - 1 = \Phi_1\Phi_2\Phi_4 = (x - 1)(x + 1)(x^2 + 1).$$

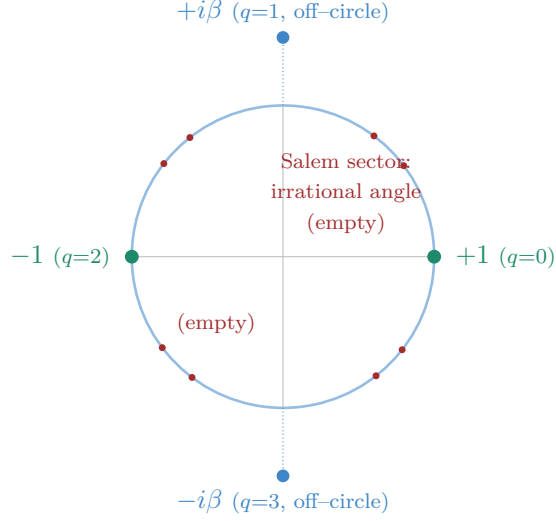


Figure 1: The $\mathbb{Z}/4\mathbb{Z}$ angle charge. The four directions $\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$ are the roots of the content polynomial $x^4 - 1$. Realized on-circle: ± 1 (charges 0, 2, **green**). The $\pm i$ sector (charges 1, 3) is realized only *off* the circle, as K 's place $\pm i\beta$ (**blue**) — the full group is in the charge, only $\mathbb{Z}/2\mathbb{Z}$ on the circle. Every irrational angle (**red**) is the Salem sector, off the charge lattice and empty.

The realized on-circle roots are $\{+1, -1\}$ ($\varphi \otimes \varphi$ contributes -1 ; $\varphi^4 \otimes \varphi^4$ contributes $+1$); the charge- $\{1, 3\}$ sector $\Phi_4 = x^2 + 1$ is realized only off the circle, by K 's place $\pm i\beta$ at modulus $\beta = 2.4195$. **[COMPUTED]**

An order- n charge produces $x^n - 1$ as the polynomial whose root set is the allowed on-circle alphabet; here $n = 4$. The split of $x^4 - 1$ across the circle (Figure 1) is the subtle part, and it carries the weight of the whole claim, so we make it explicit.

Remark 3.2 (The charge is in the algebra, not on the circle). The full $\mathbb{Z}/4\mathbb{Z}$ lives in the *charge*; only $\mathbb{Z}/2\mathbb{Z} = \{\pm 1\}$ is realized on the *circle*. The reason is K : it carries charges $\{1, 3\}$, but its imaginary place has modulus $\beta = 2.4195 \neq 1$, so the $\pm i$ sector is present in the algebra and absent from the unit circle. This is exactly why $x^4 - 1$ is the correct content polynomial and $x^2 - 1$ is not. The roots of $x^4 - 1$ are the four positions *where each charge would sit on the circle if K 's place were on it* — the $\pm i$ are the charge- $\{1, 3\}$ marks the algebra carries but the circle never shows. So $x^4 - 1$ records the charge *group*; $x^2 - 1$ would record only its on-circle *shadow*. A reader who expected the content to be $x^2 - 1$ from the realized on-circle roots alone would be undercounting the charge; the off-circle K -sector is the missing half, and recording it is what keeps the claim from overstating (the grading is genuinely $\mathbb{Z}/4\mathbb{Z}$) and from understating (only $\mathbb{Z}/2\mathbb{Z}$ is on the circle).

This off-circle placement of the $\pm i$ sector is also the angle confinement that forbids a Salem conjugate: a Salem needs modulus one at an irrational angle, while the only generator reaching argument $\pm \frac{\pi}{2}$ puts it at modulus $\beta \neq 1$ (companion paper, the metric face).

4 Object III: the gap (the cost floor φ)

A conserved charge that empties a sector turns that emptiness into a quantitative bound on the realized complement. Here the bound is the cost floor.

Proposition 4.1 (The base gap). *No monic integer quadratic $x^2 + bx + c$ has Mahler measure in the open interval $(1, \varphi)$; the smallest measure exceeding 1 is exactly φ , at $x^2 - x - 1$. [COMPUTED] (enumeration over $|b|, |c| \leq 6$: band $(1, \varphi)$ empty; realized values begin $\varphi, 2, 1+\sqrt{2}, \varphi^2, 1+\sqrt{3}, 3, \dots$)*

Corollary 4.2 (The generated floor). *Since the operators map $\{1\} \cup [\varphi, \infty)$ into itself ($\oplus$ multiplies measures, squaring squares them, $\otimes$ composes; companion paper, Lem. 6.2) and no Salem number is emitted (the off-charge sector is empty), the Mahler spectrum of $\mathcal{S}$ is $\{1\} \cup [\varphi, \infty)$, and the description-length cost carries the uniform floor $\lambda \log \varphi > 0$ at every size. [FORCED]*

This is the void as a bound. The forbidden band $(1, \varphi)$ in measure space is the height image of the off-charge sector — a sub- φ Salem is the only thing that could populate it — and its emptiness is the floor. The number φ appears as the right edge of the gap: the smallest realizable growth above the trivial $M = 1$.

5 Object IV: the graded normal form

The grading induces a canonical factorization of every emittable object along the charge.

Proposition 5.1 (Graded normal form). *Every object of $\mathcal{S}$ factors over $\mathbb{Z}$ as*

$$P = \underbrace{C(x)}_{\text{on-circle, } C|x^4-1} \cdot \underbrace{G(x)}_{\text{off-circle, } M(G)=M(P) \geq \varphi},$$

with C a product of Φ_1, Φ_2, Φ_4 and G splitting further by charge into a real (charges 0, 2) grow factor and, when K is present, an imaginary-axis (charges 1, 3) factor. [COMPUTED]

The three diagnostic cases:

object	graded factorization	sectors
$\varphi \otimes \varphi$	$(x+1)^2(x^2-3x+1)$	Φ_2^2 (on-circle, $q=2$) $\times$ grow $M=\varphi^2$
$\varphi^4 \otimes \varphi^4$	$(x-1)^2(x^2-47x+1)$	Φ_1^2 (on-circle, $q=0$) $\times$ grow $M=46.98$
$K \otimes K$	$(x^2+5)^4(x^2-5x-5)^2(x^2+5x-5)^2$	imaginary $q=1, 3 \times$ grow $M=5.854$

The first two are pure cyclotomic-times-grow; the third shows the K -sector (x^2+5) (roots $\pm i\sqrt{5}$, charge $\pm \frac{\pi}{2}$, off circle) as the third graded piece. In all cases the on-circle part divides x^4-1 and the growth lives entirely in the off-circle real factor. The normal form is the structural reason the slot is empty: a Salem is *irreducible* with on-circle conjugates at irrational angle, but the charge forces every emittable object into this *reducible*, charge-sorted shape, in which on-circle roots are roots of unity. Irreducibility and the grading are incompatible.

6 Object V: the conserved current

Objects I–IV are static structure. Object V is the same charge read dynamically, and we are explicit about what that adds and what it does not.

Proposition 6.1 (Block-diagonal evolution). *Any evolution of $\mathcal{S}$ built from the operators is block-diagonal with respect to the $\mathbb{Z}/4\mathbb{Z}$ grading: it cannot carry amplitude from one charge sector to another. Hence the charge is a constant of motion, and no admissible dynamics — iteration of $\oplus, \otimes, (\cdot)^2$, or the monodromy of any closed parameter loop — transitions into the off-charge sector. [FORCED]*

As a statement, Proposition 6.1 is the conservation law of Theorem 2.2 restated in dynamical language; it is thinner than Objects I–IV, and we say so plainly. What it adds beyond restatement is one verified consequence.

Proposition 6.2 (Clean radial growth). *Under iteration of the operators, every Mahler measure stays in $\{1\} \cup [\varphi, \infty)$ — never the forbidden band $(1, \varphi)$ — and every on-circle root stays a root of unity. [COMPUTED] (two generations deep: measures $\varphi, \varphi^2, \varphi^4, 46.98, 76.63, 122.99, 8049.92, \dots$, on-circle parts roots of unity throughout)*

So the floor of Object III is not only a static fact about the spectrum but a dynamical invariant of the orbit: because the on-circle sector is frozen (roots of unity, measure 1) and growth is confined to the off-circle grow factor, the measure evolves multiplicatively within $\{1\} \cup [\varphi, \infty)$ at every step. The growth is clean — radial, charge-preserving, floor-respecting. The remaining dynamical content, the *boundary* flow that carries a would-be occupant presented at the sector edge to a totally real Perron point with a $\sqrt{5}$ limit, is the trace redirection $\beta \mapsto \beta + \beta^{-1}$; that is the subject of the companion paper *The Occupant of the Salem Slot* and is not reproduced here.

7 Minimality: the floor is derived, not posited

Each generated object is the smallest of its kind, because each input is forced to its minimal value — and the output that matters most is the floor.

Proposition 7.1 (The minimal chain). *The generating data are forced and minimal in sequence:*

1. **Degree two** is the unique degree whose self-action realizes a ternary decision: $\#channels(d) = d^2 - d + 1 = 3$ iff $d = 2$, with the centralizer the inseparable 0-channel. (Companion paper, ternary lock.)
2. **Order four** is the charge: $\frac{\pi}{2}\mathbb{Z} \cong \mathbb{Z}/4\mathbb{Z}$ is the coarsest quantization carrying a “fourth direction” distinct from the three real channels, and its $\pm i$ generator enters through the quartic field $\mathbb{Q}(5^{1/4})$ — whose splitting field contains i — i.e. through the same K that is the construction’s one Lorentzian seed.
3. **The floor** φ is the output: the smallest Perron number, the smallest Mahler measure of a non-cyclotomic integer quadratic, the right edge of the minimal gap.

[FORCED]

Read in the forced direction, the chain says something specific about the floor, and it is worth stating without ornament. The framework posits one thing about growth decisions — that they are ternary — and ternary arity forces degree two (Proposition 7.1(1)). The cost floor φ is then *not an input*: it is the smallest realizable growth above the trivial, forced out the far end of the construction (Proposition 4.1). φ was not chosen for being golden; degree-two decisions were chosen, and φ is what they produce. The floor is a consequence of the decision arity, not an assumption adjoined to it.

That is the whole content of the minimality result: in this construction the golden ratio is *derived*. The chain $\text{degree } 2 \Rightarrow \mathbb{Z}/4\mathbb{Z} \Rightarrow \varphi$ is three forced minima producing three forced outputs — the smallest grading, the smallest content polynomial $x^4 - 1$, the smallest gap — and the appearance of φ at the end is the theorem, not a hypothesis. This is the load-bearing result of the paper, and its weight is exactly that and no more: a floor that the decision structure forces, in one fully computable instance.

8 Scope: a universal pattern, a local instance

The equivalence of §1 is domain–universal; the five objects of §§2–6 are local. Both halves matter, and conflating them is the error to avoid.

domain		conserved charge	empty sector	what the emptiness generates
fermions (QM)		wavefunction antisymmetry	doubly–occupied states	shell structure, degeneracy pressure
charge superselection (QFT)		total charge	cross–sector coherences	block structure of observables
type theory		subject reduction	ill–typed states	“illegal states unrepresentable,” optimizations
emission (here)	algebra	angle charge $\mathbb{Z}/4\mathbb{Z}$	the Salem slot	$x^4 - 1$, the floor φ , the graded normal form

The table is to be read *down each column*, never causally across rows. What is shared is the *skeleton*: a conserved charge makes a sector kinematically empty, and the emptiness organizes the realized complement (a grading, a content alphabet, a gap, a normal form, a conserved current). Each row’s output is that row’s own — the shell structure is a fact about antisymmetric fermions, $x^4 - 1$ is a fact about the angle charge — and no row’s output is derived from, or explains, another’s. In particular nothing here is a claim that the angle charge $\mathbb{Z}/4\mathbb{Z}$ explains Pauli exclusion, or that the periodic table and $x^4 - 1$ share more than the structural form of “forbidden sector organizes the realized complement.” Pauli exclusion is the most familiar instance of the *skeleton* — an antisymmetry charge whose forbidden sector has zero amplitude, never violated — and it is invoked only at that structural level.

What does not transfer is the content. The five objects this charge generates — $x^4 - 1$, the floor φ , the graded factorization, the trace flow with its $\sqrt{5}$ limit — are theorems of *this* number theory, not exported engines for other fields. The only genuinely portable mathematics beyond the pattern is the *local* differential structure: a two–to–one cover with a fold has a square–root branch point generically (Chebyshev maps, modular curves, mapping–class monodromy), so the $\sqrt{t - 2}$ edge and the $\beta \leftrightarrow \beta^{-1}$ monodromy recur widely; but $\sqrt{5} = \varphi + \varphi^{-1}$, the identification of the floor’s image with the grow generator, is this construction’s own. The honest statement is precise: the substrate (kinematic emptiness $\equiv$ conserved charge $\equiv$ superselection) is one of the most generative patterns there is, and the Salem slot is a clean, minimal, fully computable realization of it — a generator of its own domain’s structure, exhibiting in the small what the pattern does at large.

9 Epistemic ledger

Claim	Status	Backing
Kinematic vs. dynamic emptiness; the substrate equivalence	[DECLARED]	Def. 1.1, §1
Object I — angle charge is $\mathbb{Z}/4\mathbb{Z}$, conserved, fully realized (via K)	[COMPUTED]	Thm. 2.2
Operators act as add ($\otimes$), double (squaring), union ($\oplus$)	[COMPUTED]	Thm. 2.2
Object II — content polynomial $x^4 - 1 = \Phi_1 \Phi_2 \Phi_4$	[COMPUTED]	Prop. 3.1
Full $\mathbb{Z}/4\mathbb{Z}$ in the charge; only $\mathbb{Z}/2\mathbb{Z} = \{\pm 1\}$ on the circle	[COMPUTED]	Rem. 3.2
Object III — no integer quadratic with $M \in (1, \varphi)$; floor φ	[COMPUTED]	Prop. 4.1, Cor. 4.2
Object IV — graded normal form, on-circle $ x^4 - 1 \times \text{grow } M \geq \varphi$	[COMPUTED]	Prop. 5.1
Object V — block-diagonal evolution (conservation restated)	[FORCED]	Prop. 6.1
Object V — clean radial growth: orbit measures in $\{1\} \cup [\varphi, \infty)$	[COMPUTED]	Prop. 6.2
Minimal chain degree $2 \Rightarrow \mathbb{Z}/4\mathbb{Z} \Rightarrow \varphi$; floor derived not posited	[FORCED]	Prop. 7.1
Pattern universal (Pauli, superselection, types); content local	[DECLARED]	§8
$\sqrt{5} = \varphi + \varphi^{-1}$ local; $\sqrt{\cdot}$ branch & monodromy portable	[FORCED]	§8

Summary. A kinematic void — a sector emptied by a conserved charge rather than guarded by a barrier — is a generator. On the minimal instance, the angle charge $\mathbb{Z}/4\mathbb{Z}$ of the Salem slot produces five canonical objects: the grading (I), the content polynomial $x^4 - 1$ (II), the cost floor φ (III), the graded cyclotomic-times-grow normal form (IV), and the conserved current whose dynamical content is clean radial growth (V). The floor is the result that carries weight: degree two is forced by ternary arity, and φ falls out the far end as the minimum realizable growth, so the golden ratio is derived from the decision structure rather than posited. The pattern beneath — kinematic emptiness $\equiv$ conserved charge $\equiv$ superselection — is the same skeleton that, in each domain, does that domain's own work: for fermions the shell structure, for type systems their correctness, here the five objects; the skeleton is shared, the outputs are not exported. The void does not merely sit there refusing to be filled. Refusing to be filled, in a conserved way, is the act of generation.